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MILITARY SPECIFICATION

GENERATOR, AIMS PRETRIGGER SG-841( )/UPX

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pretrigger generator which provides all necessary timing signals for shipboard AIMS system operation.

1.2 Operating modes. The pretrigger generator covered by this specification operates in either of the following two modes:

Mode A: Operation with radar sets utilizing a nonstaggered or nonjittered pulse repetition frequency (PRF).
Mode B: Operation with radars utilizing a jittered PRF.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.i. (High-impact), Shipboard Machinery, Equipment And Systems, Requirements For (Navy)
MIL-E-4158	Electronic Equipment, Ground, General Requirements For
MIL-E-16400[1/]	Electronic Equipment, Naval Ship And Shore: General Specification
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-C-39012/21	Connectors, Coaxial, Radio Frequency (Series BNC (Uncabled) Receptacles, Female, Jam Nut Mounted, Class II)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5840

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definition Of And Basic Requirements For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type 1- Environmental And Type 1i - Internally Exited)
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution

1/ Revision F applies.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with requirements of MIL-E-16400 except as hereinafter specified.

3.1.1 Subassemblies and circuit-card assemblies. Subassemblies and circuit-card assemblies which have been developed and tested under the requirements of MIL-E-4158 and which use materials and finishes in accordance with MIL-E-4158 will be acceptable for use in this equipment without need for redesign specifically to meet the requirements of MIL-E-16400.

3.1.2 First article. Prior to beginning production, a sample shall be tested as specified in 4.3.

3.2 Design.

3.2.1 Equipment composition. The equipment shall consist of solid state circuitry mounted on printed circuit cards, mounted in an enclosure which is dripproof in accordance with MIL-STD-108.

3.2.1.1 Printed circuit cards. All printed circuit cards used shall be designed and manufactured in accordance with MIL-STD-275. Each type of card used in the equipment shall be marked with an identifying number to distinguish it from any other type.

3.2.1.2 Operating configurations. The pretrigger generator covered by this specification shall be capable of operating in either mode A or mode B as described in 1.2. The equipment may convert from mode A operation to mode B operation by an interchange of modules within the equipment. Modules which are used for mode A operation but are not used for mode B operation shall be stored within the equipment during mode B operation, and modules which are used for mode B operation but are not used for mode A operation shall be stored within the equipment during mode A operation.

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3.2.2 Electrical design. The equipment shall be capable of operating from a power source whose voltage and frequency characteristics are as follows:

a. Voltage: 115 VAC ±10 percent single phase
b. Frequency: 50, 60 and 400 Hz ±10 percent

3.2.2.1 Overload protection. Overload protection is required. Wherever possible overload protection shall be provided by circuit breakers in lieu of fuses. Regardless of the type of protection used, all equipment shall have an indicator on the front panel, near the primary power indicator, which shows when an overload has caused the protective device to function.

3.2.2.2 External connectors. All signal connectors shall be type C, and all power and control connectors shall be chosen from the AN or MS series. All connectors shall be on the rear panel.

3.2.2.2.1 DC isolation. The video signal inputs shall be capacitively coupled to insure proper system operation with signals which have a DC reference other than ground. This requirement shall not be waived unless the contractor demonstrates to the procuring activity and obtains approval of a design technique which is immune to such input signals.

3.2.2.2.2 Output reference level. The video signal outputs shall be referenced to ground minus 0.1 volt plus 0.2 volt unless otherwise specified herein.

3.2.2.3 Electromagnetic interference emission, susceptibility, and equipments and systems self-compatibility. The equipment shall meet the requirements of MIL-STD-461 for Class IC equipment except that CS03, CS04, CS05, (T) CS07 and CS08 are not required.

3.2.3 Mechanical design. The mechanical design of the equipment shall conform to the requirements of 3.2.3.1 through 3.2.3.6.

3.2.3.1 Size. The maximum dimensions (including mounting provisions) of the equipment shall be as follows:

a. Height: 12.7 cm (5 inches)
b. Width: 44.45 cm (17 1/2 inches)
c. Depth: 30.48 cm (12 inches)

3.2.3.2 Weight. The weight of the equipment shall not exceed 9.07 kg (20 pounds).

3.2.3.3 Mounting. The equipment shall be designed for mounting to a flat surface without resilient mounts. Provisions shall be included for mounting the bottom surface of the equipment to this flat surface.

3.2.3.4 Component chassis. The chassis assembly shall be composed of a front panel and a component chassis. All circuit boards and subassemblies shall be accessible when the chassis is withdrawn from the equipment case. Access to the equipment interior shall be accomplished by releasing no more than six securing devices located on the front panel surface only.

3.2.3.5 Equipment case. The case shall form the top, bottom, and two lateral sides of a rectangular box comprising the equipment enclosure. The case shall have provisions for direct mounting to an equipment shelf by the bottom surface only.

3.2.3.6 Rear panel. The rear panel of the equipment shall be a removable part of the enclosure and not a part of the equipment chassis. All signal input and output connectors shall be installed on the removable rear panel except for test signal outputs (see 3.4.3.2) which shall be on the front panel.

3.2.3.7 Enclosure. The equipment shall be mounted by devices secured to the base of the enclosure. The equipment shall be designed to permit ready access to internal test points and replaceable modules via the front of the equipment. MIL-STD-454, Requirement 36 shall apply.

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3.2.4 Controls, indicators, and panel layout.

3.2.4.1 Tactile identification. Tactile identification of controls is not required.

3.2.4.2 Classification of controls. All controls shall be classified as continually required except those used primarily in initial installation and adjustment or for maintenance adjustment and alignment.

3.2.5 Leakage current (equipment). When practicable, the leakage current of the equipment shall not exceed 5 mA to ground, whether or not such equipment contains radio interference filters or capacitors. Should the equipment have an unavoidable leakage current in excess of 5 mA, a warning plate shall be attached to the front panel of each equipment reading:

DANGER - Do not energize this equipment unless frame and all exposed metal parts are grounded.

3.3 Reliability.

3.3.1 Quantitative reliability. The upper test mean-time-between-failure (MTBF) (θ_0 as defined in MIL-STD-781) for the pretrigger generator shall be 6000 hours. The lower test MTBF (θ_1) shall be 2000 hours. Failure is the inability to meet any of the performance control or monitoring functions specified herein.

3.3.2 Equipment conditioning. The equipment shall be capable of passing a temperature equipment conditioning test, with the environment temperature equal to or greater than 50°C, with the equipment operating in the worst case mode for a continuous operating time of 72 hours of failure-free time. Failures occurring during this 72 hour period shall be repaired and the burn-in resumed until 72 hours of failure-free time is achieved. No adjustments of the equipment, other than normal operating mode changes, shall be allowed during this test.

3.4 Maintainability.

3.4.1 Quantitative maintainability requirements. The pretrigger generator shall have an equipment repair time (ERT) of 15 minutes when corrective maintenance by module or circuit card is performed. ERT includes localization, isolation, disassembly, interchange, reassembly, alignment and checkout for all subassemblies and parts.

3.4.2 Maintainability demonstration. The maintainability requirements for the pretrigger generator shall be demonstrated in accordance with 4.6.

3.4.3 Service and access. The equipment design shall permit easy and ready access to internal parts, test points, and adjustments. Removal and replacement of subassemblies shall not require the removal or displacement of other subassemblies.

3.4.3.1 Accessibility of internal test points and replaceable modules. Equipment procured under this specification shall be designed to permit ready access to internal test points and replaceable modules via the front of the equipment in accordance with Requirement 36 of MIL-STD-454.

3.4.3.1.1 Captive fasteners. Loosening of captive fasteners on the front panel shall provide for withdrawal of the equipment from the case, for maintenance.

3.4.3.1.2 Slides, guides, safety interlocks, and bypass switches. Slides or guides with safety stops to prevent accidental complete removal shall be employed. Safety interlocks and bypass switches are required.

3.4.3.1.3 Internal power and signal cables. The internal power and signal cables shall be of sufficient length to allow full extension of the unit from the case. Provisions shall be supplied to support all cables coming from the rear panel to the removable component chassis, such that these cables sustain no damage.

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3.4.3.2 Accessibility of external test points. Test points shall be provided on the front panel of the equipment. Signals available shall be all of the equipment input/output signals. Type BNC UG-625 connectors in accordance with MIL-C-39012/21 shall be mounted on the front panel of the equipment for measurement of these signals.

3.4.4 Reversibility. The equipment shall incorporate features such that it is mechanically or electrically impossible to install incorrectly its modules, circuit card assemblies, and sub-assemblies, except as required for compliance with the interchange of modules permitted (see 3.2.1.2).

3.5 Maintenance.

3.5.1 Test points and test features. Test points and test features are required and shall comply with 3.5.1.1 through 3.5.1.3.

3.5.1.1 External test points. Test points shall be provided on the front panel to allow rapid, accurate isolation of malfunction to the defective equipment. Mechanical design of the front panel shall be as specified in 3.4.3.2.

3.5.1.2 Internal test points. Internal test points shall be provided on each circuit board or module. Observation of these test points will identify a defective board or module immediately. Test points for printed circuit cards shall be brought to the top of the card. The test points shall be of the probe type, suitable for insertion of a standard, flexible tip probe.

3.5.1.3 Extender board. A printed circuit extender board shall be provided. One extender shall accommodate all printed circuit boards of the equipment. It shall be stored in the card compartment when not in use.

3.6 Performance characteristics and detail requirements.

3.6.1 Performance characteristics.

3.6.1.1 Input signals. The equipment shall operate with input signals having characteristics as described in 3.6.1.1.1 and 3.6.1.1.2.

3.6.1.1.1 Radar trigger.

a.	Polarity:	Positive
b.	Amplitude:	5 to 55 volts
c.	Duration:	0.30 to 25.0 microseconds (µsec)
d.	Rise Time:	Up to 25 percent of pulse duration, or 0.04 µsec per volt times pulse amplitude, whichever is smaller
e.	Fall Time:	1 to 5 times pulse rise time
f.	Source Impedance:	75 ohms, resistive, ±5 ohms
g.	PRF:	15 - 3000 pps mode A 15 - 2000 pps mode B

3.6.1.1.2 Input power. The equipment shall require no more than 25 volt-amperes primary power. The equipment's power factor shall be no less than 0.85.

3.6.1.2 Output signals. The equipment shall produce output signals with characteristics as specified in 3.6.1.2.1 through 3.6.1.2.3.

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3.6.1.2.1 SIF pretrigger. The SIF pretrigger shall be as follows:

a.	Polarity:	Positive
b.	Amplitude:	20 ±5 volts
c.	Noise Level:	0 ±0.5 volts
d.	Duration:	1.0 ±0.2 µsec
e.	Rise Time:	0.1 µsec, maximum, 0.02 minimum
f.	Fall Time:	0.1 µsec, maximum, 0.02 minimum
g.	Termination Impedance:	75 ±5.0 ohms, resistive

3.6.1.2.2 Other trigger outputs. The following additional output triggers shall be produced with characteristics identical to the SIF pretrigger as described in 3.6.1.2.1.

a. Processor start trigger
b. Mode 4 pretrigger
c. Radar trigger (except for mode A, external triggering) output shall have two selectable amplitudes, 20 ±5 volts and 45 ±10 volts. Selection shall be made by internal solderless link adjustment. All other characteristics shall be in accordance with 3.6.1.2.1.

3.6.1.2.3 Remote fault indication. The remote fault indication signal shall provide a suitable ground for a 6.3 VAC, 200 milliampere lamp.

3.6.2 Detail requirements.

3.6.2.1 Electrical connectors, controls, indicators and mechanical considerations. The equipment shall be provided with the connectors, controls and indicators required by 3.6.2.1.1 through 3.6.2.1.6 and shall conform to the mechanical design requirements of 3.2.3.

3.6.2.1.1 Power connector. The power connector (rear panel) shall be an MS or AN (male) connector with three pins to accept 115 VAC, single-phase:

a. Pin A 115 VAC
b. Pin B 115 VAC return
c. Pin C AC system ground

3.6.2.1.2 Signal connectors. The equipment shall be provided with one Type C coaxial connector (rear panel) for each of the following signals:

a. Radar trigger input
b. SIF pretrigger output
c. Processor start trigger output
d. Mode 4 pretrigger output
e. Radar trigger output

3.6.2.1.3 Remote fault indicator connector. The equipment shall be provided with a rear panel remote fault indicator connector (see 3.6.1.2.3). This connector shall be an MS or AN (female) connector with five pins.

3.6.2.1.4 Mating connectors. Mating connectors shall be provided with the equipment for the rear panel power connector (see 3.6.2.1.1), signal connectors (see 3.6.2.1.2), and the remote fault indicator connector (see 3.6.2.1.3).

3.6.2.1.5 Front panel controls. The equipment shall be provided with the following front panel controls as a minimum:

a. Power on-off (toggle switch)
b. Int-Ext (toggle switch)
c. Pretrigger timing (see 3.6.3.1.1)
d. Pulse repetition frequency (PRF) select (see 3.6.3.1.4)

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3.6.2.1.6 Front panel indicators. The equipment shall be provided with the following front panel indicators as a minimum:

a. Power on
b. Fault indication (see 3.6.3.1.5)
c. Misalignment indication (see 3.6.3.2.1)

3.6.3 Functional characteristics. Switching between modes A and B may be accomplished by interchange of modules.

3.6.3.1 Common functional characteristics. Functional characteristics common to both operational modes are described in 3.6.3.1.1 through 3.6.3.1.5.

3.6.3.1.1 Relative timing. The timing of each output signal, as specified in 3.6.1.2.2 shall be independently adjustable from zero to minus 600 µsec relative to Radar Trigger, or t_o. The SIF pretrigger shall be adjustable from zero to minus 99 µsec relative to t_o (t_o is defined for modes A and B in subsequent paragraphs). These adjustments shall be accomplished in increments no greater than one µsec by front panel switches. All internal timing functions of the equipment shall use a common crystal-controlled clock and digital counting circuits. The absolute accuracy of all time intervals between any pretrigger and the following t_o shall be within ±0.4 µsec of the selected interval. This accuracy shall be maintained without readjustment while undergoing environmental tests during a 24-hour period.

3.6.3.1.2 Input PRF range. The equipment shall accept Radar Trigger inputs over the range of 15 to 3000 pulses per second (pps).

3.6.3.1.3 Countdown. When the input PRF exceeds a selected number between 230 and 410 pps, all outputs except the Radar Trigger output shall be counted down to the largest integral sub-multiple of the input PRF which is smaller than the selected number. The selected number shall be chosen by internal adjustment in increments not greater than 30 pps. The Radar Trigger output PRF shall be identical to the input PRF.

3.6.3.1.4 Free-run operation. The equipment shall generate all output signals at a PRF selected by a front panel switch, independently of any input signal(s). The selection of internal PRF shall be accomplished in steps not greater than 10 µsec of pulse repetition time (PRT) over a range of 200 pps to 400 pps. The jitter on the internal PRF shall not exceed ±0.2 µsec. A radar trigger output (t_o) shall be provided.

3.6.3.1.5 Fault indication. A fault indicator shall be provided for each of the pretrigger outputs. Each indicator shall be illuminated whenever its pretrigger output is not present.

3.6.3.2 Mode A operation. In mode A, the input trigger shall occur at a single, constant PRF. In this mode, the equipment shall count the time interval between incoming pulses, and shall provide output signals prior to the expected input triggers at the selected relative timing. For mode A, t_o is defined as the 50 percent amplitude point on the leading edge of the input trigger.

3.6.3.2.1 Misalignment indication. A lamp shall be illuminated whenever the time difference between the expected input trigger and the actual input trigger exceeds ±1 µsec. Separate indicators shall be provided for expected trigger before actual trigger and vice versa. If tracking is accomplished by a period to period method that automatically maintains alignment, misalignment indicators are not required.

3.6.3.2.2 Input PRF drift. The absolute accuracy of each pretrigger time (see 3.6.3.1.1) shall be maintained without readjustment for a long term drift of the input PRT of at least ±5 µsec. Long term drift shall be defined as a linear variation over a 30-minute period.

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3.6.3.2.3 *Timing jitter.* The maximum jitter between any pretrigger and the following t_o shall not exceed ±0.2 µsec. This requirement shall apply for a stable input trigger. Provisions shall be incorporated for operation in a degraded mode with input triggers with a jitter of up to ±2.0 µsec. In this case, all timing accuracy and timing jitter tolerances shall be increased by an amount equal to the input trigger jitter.

3.6.3.2.4 External operation. When the equipment is in mode A, an external trigger input shall be directly connected to the radar trigger output. The output pulse characteristics shall then be identical to the input pulse characteristics.

3.6.3.3 Mode B operation. When the equipment is configured for mode B operation, it will effectively revert to a time-delay device. That is, the equipment shall produce its outputs at relative delayed times after receipt of each input trigger, until the end of a total delay interval, whereupon it will produce a radar trigger output. In the mode B, t_o is defined as the 50 percent point on the leading edge of the radar trigger output signal.

3.6.3.3.1 Fail-safe feature. When mode B operation is in use, any failure recognized by the fault indication circuits shall cause the radar trigger input to be connected directly to the radar trigger output. The equipment shall be so designed that a failure in the input power will also cause the input radar trigger to be fed through the equipment to the radar trigger output. This will insure continued operation of the associated radar set when the equipment experiences a failure.

3.6.3.3.2 Timing jitter. The maximum jitter between the input trigger and the radar trigger output (t_o) or between any pretrigger and t_o shall not exceed ±0.1 microsecond.

3.6.3.3.3 Remote fault indication. The remote fault indication signal shall be activated whenever a fault is sensed in any of the t_o (see 3.6.3.2.1).

3.6.4 Environmental conditions. The equipment shall be capable of operation under any combination of the following conditions:

a. Temperature - Class 2 of MIL-E-16400 except that the upper limit shall be 65°C for both operating and non-operating conditions.
b. Humidity, up to 95 percent relative.
c. Shock, vibration, inclination and noise.

Semiconductors shall be qualified for operation over the full military temperature range.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, (a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; (b) Government product inspection to measure quality of the product to be offered for acceptance; (c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The methods of examination and testing of the equipment fall within the following classifications:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (see 4.4.1)
   2. Production control inspection (see 4.4.2)
   3. Environmental test (see 4.4.3)
c. Reliability demonstration (see 4.5)
d. Maintainability demonstration (see 4.6)
e. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2) two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall consist of the listed preproduction examinations and tests of MIL-E-16400 and shall be performed in accordance with the inspection procedures of MIL-E-16400 except as specified herein.

4.3.1 Water cooling, reliability, and accelerated life. The water cooling, reliability, and accelerated life tests of MIL-E-16400 are not required.

4.3.2 Leakage current (equipment). Leakage current (see 3.2.5) shall be measured in the following manner. The equipment shall be placed on an insulated surface and test connections made as shown in the Test Diagram for Leakage Current (FIGURE 1). The ground connection between the equipment and the safety ground conductor shall be opened. The equipment shall be operating with nominal input voltage and frequency. The probe shall be connected to the equipment enclosure and the leakage current measured for every combination of switch positions available in the test diagram. All parts not electrically common with the equipment enclosure shall also be tested for leakage current. If the equipment has more than one mode of operation, such as transmit and receive, it shall be tested in each mode.

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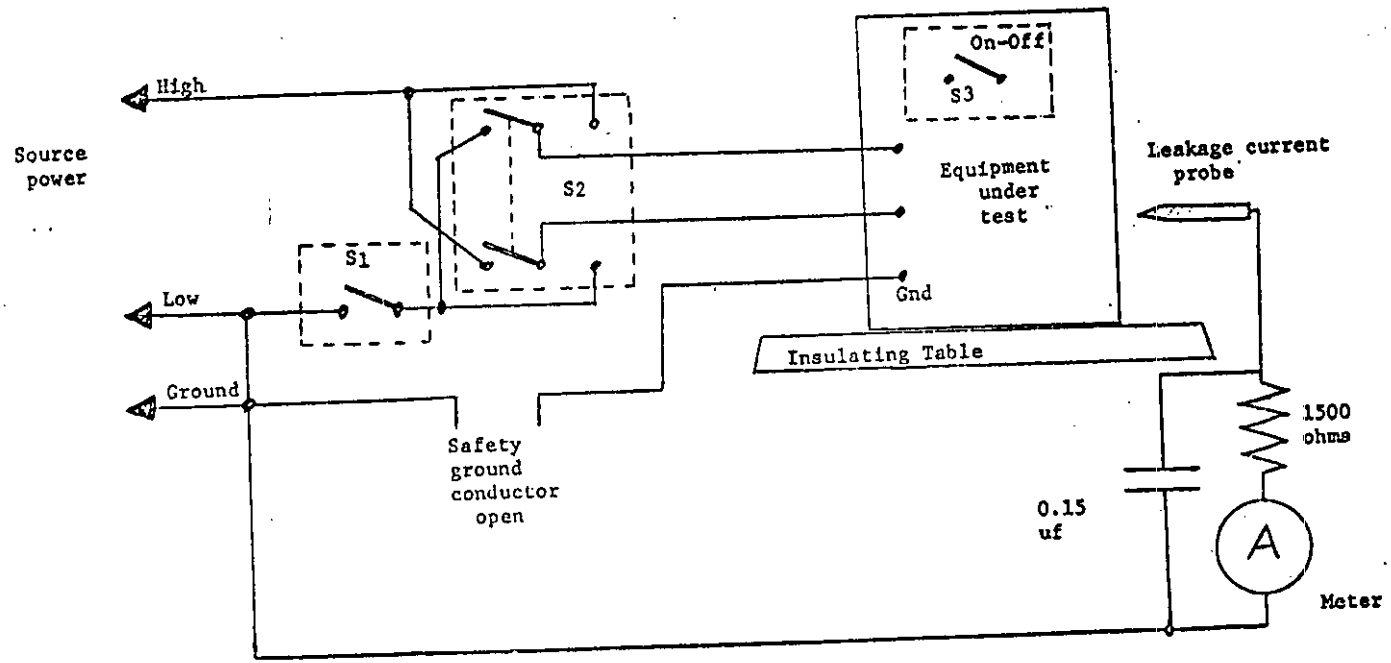


FIGURE 1. Test diagram for Leakage current.

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4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the examinations and tests of MIL-E-16400 and shall be conducted in accordance with the inspection procedures of MIL-E-16400. The operating test will test each electrical function of the equipment.

4.4.2 Production control inspection. Production control inspection including sampling, shall be limited to TABLE I and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control tests shall be performed in accordance with the production control inspection procedures of MIL-E-16400. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE I. Production control inspection.

Inspection	Requirement paragraph	AQL%
Weights and dimensions	3.2.3.1, 3.2.3.2	10
Supply line voltage and frequency	3.2.2	10
Heat test	3.6.4	10
Power	3.6.1.1.2	10
Controls and control circuits	3.2.4	10
Enclosure	3.2.1	10
Leakage current (equipment)	3.2.5	10

4.4.2.1 Leakage current inspection. Leakage current (equipment) inspection shall be performed as specified in 4.3.2.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental tests. Environmental tests shall be as listed in TABLE II and shall be performed on units which have been subjected to and passed production inspection. Environmental tests shall be performed in accordance with the environmental test procedures of MIL-E-16400.

TABLE II. Environmental tests.

Inspection	Requirement
Temperature	3.6.4
Humidity	3.6.4
Shock, vibration, inclination and noise	3.6.4
Salt spray test	3.6.4

4.4.3.1 Environmental test samples. Samples for the environmental inspection shall be randomly selected in accordance with the following:

a. One sample from the first 10 production units fabricated
b. One sample from each successive 75 units produced, or fraction thereof

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4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the tests specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units which were manufactured with the same conditions, materials and processes. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental sample units. Unless otherwise specified in the contract (see 6.2), sample units which have been subjected to and passed production control or environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.5 Reliability demonstration. The reliability of the pretrigger generator shall be demonstrated and reported in accordance with MIL-STD-781.

4.5.1 Reliability testing. A sample of at least four production units shall be subjected to production reliability acceptance testing in accordance with Test Plan V C of MIL-STD-781 to demonstrate the lower test MTBF. Environmental and operational duty cycles shall be as specified herein.

4.5.1.1 Temperature. Testing shall be performed at 50° +5 -0° C.

4.5.1.2 Vibration. The equipment shall be vibrated at 2.2G ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. Vibration shall be normal to the plane of the majority of the circuit boards or cards.

4.5.1.3 Operating duty cycle. The equipment shall be operating with inputs which simulate actual operational use. Equipment shall be turned on and let temperature stabilize, hold for three hours, then turn off and let equipment temperature stabilize. This cycle shall continue throughout the test.

4.5.1.4 Input voltage. Input voltage to test samples shall be nominal specified voltage +5, -2 percent.

4.5.1.5 Moisture. Once per test day, moisture shall be injected into the test chamber during the stabilized hot temperature period. The moisture level need not be strictly controlled, but shall be sufficient to cause visible condensation in the chamber.

4.5.2 Equipment conditioning. All deliverable equipments shall be subjected to the equipment conditioning requirement of 3.3.2 to assure stabilization of the required characteristics of the equipment, elimination of initial failures due to random defective components, and marginal operation due to component lot variations, or poor assembly processes or procedures. A performance check for proper operation shall be performed a minimum of twice every 24 hours and just prior to end of test.

4.5.3 Failure. Failure is any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements of this specification. Failure categories of MIL-STD-781 shall be used as a guide by the procuring activity in determining relevancy.

4.5.4 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer designs, corrective action shall be performed on every equipment delivered under the contract.

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4.5.5 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be resubjected to the production test (see 4.4.1) to ensure that it conforms to the requirements of this specification. After reconditioning, reinspection, and retesting,the equipment may be delivered as part of the contract quantity, provided it conforms to all terms and conditions of the contract for acceptance.

4.6 Maintainability demonstration. The maintainability requirements of 3.4.2 for the pretrigger generator shall be successfully demonstrated in accordance with Method 4 of MIL-STD-471 prior to the shipment of any production equipment.

4.7 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. This specification covers the design, manufacture, and test of the AIMS Pretrigger Generator SG-841( )/UPX for shipboard use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Marking, levels of preservation, packaging, and packing required (see 5.1)
d. When sample units which have been subjected to and passed production control or environmental test may not be accepted (see 4.4.4)

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.5	Reliability demonstration test plans	DI-R-2124
4.5.1	Reliability test reports	DI-R-2118
4.6	Maintainability demonstration plans	DI-R-2129
4.6	Maintainability demonstration report	DI-R-2130A

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation (ASPR). The first article should be a preproduction sample, as specified in 4.3. The first article should consist of two units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

Preparing activity:
NAVY-EC
(Project 5840-N184)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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